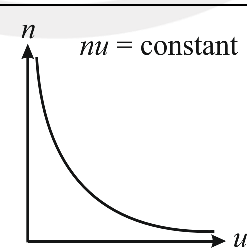
CHAPTER 1

Unit and Measurement

<table>
<tr><th>Sr. No.</th><th>Concept</th><th>Formulas</th><th>Other information</th></tr>
<tr><td>1.</td><td>Classification</td><td>Physical Quantities
→ Independent of any other quantity → Fundamental
→ Dependent on other quantities → Derived</td><td></td></tr>
<tr><td>2.</td><td>Fundamental quantities and their SI units</td><td>
<table>
<tr><th>Physical quantity</th><th>Unit</th><th>Symbol</th></tr>
<tr><td>Length</td><td>metre</td><td>m</td></tr>
<tr><td>Mass</td><td>kilogram</td><td>kg</td></tr>
<tr><td>Time</td><td>second</td><td>s</td></tr>
<tr><td>Temperature</td><td>kelvin</td><td>K</td></tr>
<tr><td>Electric current</td><td>ampere</td><td>A</td></tr>
<tr><td>Luminous intensity</td><td>candela</td><td>cd</td></tr>
<tr><td>Amount of substance</td><td>mole</td><td>mol</td></tr>
</table>
</td><td>Example of derived quantities:
(1) Area
(2) Force
(3) Density</td></tr>
<tr><td>3.</td><td>For magnitude of physical quantity numerical value and unit relation</td><td>$n \propto \frac{1}{u} \Rightarrow nu = \text{constant}$
Or, $n_1 u_1 = n_2 u_2$

</td><td>n = numerical value of the physical quantity
u = size of unit.</td></tr>
<tr><td>4.</td><td>Supplementary quantities and their SI units</td><td>Plane angle - Radian (rad)- $d\theta = \frac{ds}{r}$
Solid angle - Steradian (sr) - $d\Omega = \frac{dA}{r^2}$</td><td>r = radius
ds = arc length
dA = subtended area</td></tr>
</table>

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5.	**Dimensional representation**	$[M]$ → for mass $[L]$ → for length $[T]$ → for time $[A]$ → for electric current $[K]$ or $[\theta]$ → for thermodynamic temperature $[cd]$ → for luminous intensity [mol] → for amount of substance	
6.	**Principle of Homogeneity**	Only physical quantities having same dimension can be added or subtracted.	For example: $A + B = C - D$ The physical quantities A, B, C, and D have the same dimension.
7.	**To convert a physical quantity from one system of units to other**	If dimension of a physical quantity is represented by $[M^a L^b T^c]$ then: $n_2 = n_1\left(\frac{M_1}{M_2}\right)^a\left(\frac{L_1}{L_2}\right)^b\left(\frac{T_1}{T_2}\right)^c$	n_1 = numerical value in 1st system. n_2 = numerical value in 2nd system
8.	**Rules to find out the number of significant figures (SF)**	**Rule I:** All the non-zero digits are significant e.g. 1984 has 4 SF. **Rule II:** All the zeros between two non-zero digits are significant e.g. 10806 has 5 SF. **Rule III:** All the zeros to the left of first non-zero digit are not significant, e.g. 00108 has 3 SF. **Rule IV:** If the number is less than 1, zeros on the right of the decimal point but to the left of the first non zero digit are not significant, e.g. 0.002308 has 4 SF. **Rule V:** The trailing zeros (zeros to the right of the last non-zero digit) in a number with a decimal point are significant, e.g. 01.080 has 4 SF. **Rule VI:** The trailing zeros in a number without a decimal point may not be significant e g. 010100 has 3 SF. **Rule VII:** For any value written in notation $A \times 10^x$, the number of S.F. is determined by applying the above rules only to the value of A. For example, in $x = 12.3 = 1.23 \times 10^1 = 0.123 \times 10^2 = 0.0123 \times 10^3 = 123 \times 10^{-1}$ each term has 3 SF only.	In multiplication or division, the number of SF in the product or quotient is same as the smallest number of SF in any of the factors. e.g. $2.4 \times 3.65 = 8.8$

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9.	**Rounding off**	**Rule I:** If the digit to be rounded off is more than 5, then the preceding digit is increased by one. e.g. $6.87 \approx 6.9$ **Rule II:** If the digit to be rounded off is less than 5, then the preceding digit is left unchanged. e.g. $3.94 \approx 3.9$ **Rule III:** If the digit to be rounded off is 5 then the preceding digit is increased by one if it is odd and is left unchanged if it is even. e.g. $14.35 \approx 14.4$ and $14.45 \approx 14.4$	
10.	**Significant rules for addition and subtraction**	The answer equals the smallest number of decimal places in any of the original numbers.	$\begin{array}{r} 3.1421 \\ 0.241 \\ +\,0.09 \\ \hline 3.4731 \end{array}$ ←(has two decimal places) (Answer should be reported to two decimal places after rounding off) Answer = 3.47
11.	**Significant rules for multiplication and division**	The answer equals the smallest number of significant figures in any of the original numbers.	$\begin{array}{r} 51.028 \\ \times\,1.31 \\ \hline 66.84668 \end{array}$ ←(Three significant figures) (Answer should have three significant figures after rounding off) Answer = 66.8
12.	**Least Count (Vernier Caliper)**	Least Count = 1 MSD – 1 VSD $\text{Least Count} = 1\text{ MSD} - \frac{n-1}{n}\text{ MSD} = \frac{1\text{ MSD}}{n}$	If n VSD coincides with $(n–1)$ MSD, Then $(n–1)$ MSD $= n$ VSD $\therefore 1\text{ VSD} = \frac{n-1}{n}\text{MSD}$
13.	**Total reading (Vernier Caliper)**	Total Reading = Main scale reading + (Conciding Verneir Scale division × least count) Total reading = MSR + VSR = MSR + $n \times$ VC $n = n^{th}$ division of the vernier scale coinciding with the main scale.	MSR = Main scale reading VC = Vernier constant i.e. least count

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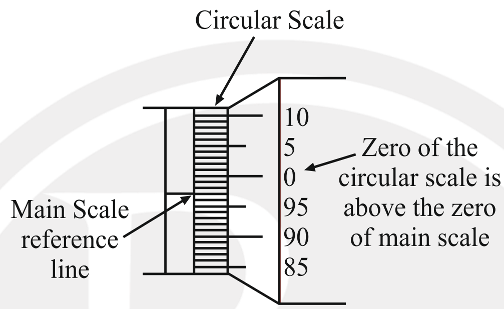
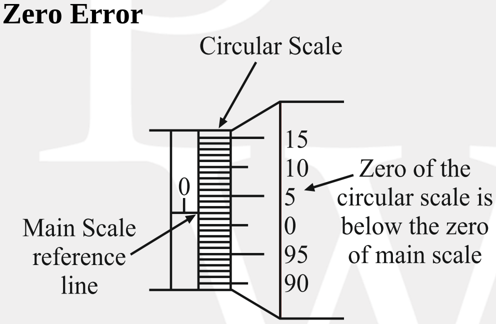
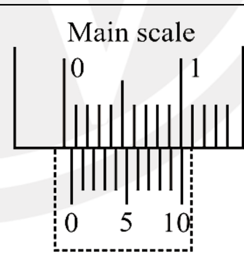
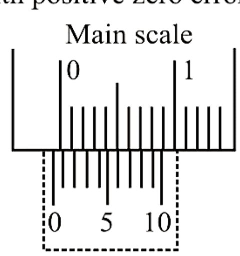
14.	**Screw Gauge**	Pitch of the screw gauge $= \dfrac{\text{Distance moved in } n\text{-rotation of circular-scale}}{\text{No. of full-rotation}}$ $\text{L.C} = \dfrac{\text{Pitch}}{\text{Total number of division on the circular scale}}$	
15.	**Total reading (Screw Gauge)**	Total Reading (T.R) = L.S.R. + C.S.R L.S.R = Linear Scale Reading = N C.S. R = Circular Scale Reading = $n \times$ L.C $\therefore$ Total reading = $N + n \times$ (L.C)	n = division of the circular scale coinciding with the linear scale line
16.	**Zero Error in Screw Gauge**	**Negative Zero Error**  **Positive Zero Error** 	Correct reading = (Reading) – (zero error)
17.	**Zero Error in Vernier Caliper**	1. If the zero of the Vernier scale is to the right of the zero of the main scale, the zero error is said to be positive.  Vernier Scale with positive zero error 2. If the zero of the Vernier scale is to the left of the zero of the main scale, the zero error is said to be negative.  Vernier Scale with negative zero error	

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<tr><td>18.</td><td>Absolute Error
(Δa)</td><td>Absolute Error = True Value – Measured Value
$\Delta a = a_T - a$
Mean absolute error is defined as
$\overline{\Delta a} = \frac{|\Delta a_1| + |\Delta a_2| + \ldots + |\Delta a_n|}{n} = \sum_{i=1}^{n} \frac{|\Delta a_i|}{n}$</td><td></td></tr>
<tr><td>19.</td><td>Percentage Error</td><td>Percentage Error = Relative Error × 100%
Percentage Error = $\frac{\text{Mean Absolute Error}}{\text{True Value}} \times 100\%$</td><td>Relative Error = (Mean Absolute Error / True Value)</td></tr>
<tr><td>20.</td><td>Calculation of percentage errors (All Cases)</td><td>
<table>
<tr><th>Operation</th><th>Formula Z</th><th>Absolute error ΔZ</th><th>Relative error ΔZ/Z</th><th>Maximum percentage error (ΔZ/Z) × 100</th></tr>
<tr><td>Sum</td><td>$A+B$</td><td>$\Delta A + \Delta B$</td><td>$\frac{\Delta A + \Delta B}{A+B}$</td><td>$\frac{\Delta A + \Delta B}{A+B} \times 100$</td></tr>
<tr><td>Difference</td><td>$A-B$</td><td>$\Delta A + \Delta B$</td><td>$\frac{\Delta A + \Delta B}{A-B}$</td><td>$\frac{\Delta A + \Delta B}{A-B} \times 100$</td></tr>
<tr><td>Multiplication</td><td>$A \times B$</td><td>$A\Delta B + B\Delta A$</td><td>$\frac{\Delta A}{A} + \frac{\Delta B}{B}$</td><td>$\left(\frac{\Delta A}{A} + \frac{\Delta B}{B}\right) \times 100$</td></tr>
<tr><td>Division</td><td>$\frac{A}{B}$</td><td>$\frac{B\Delta A + A\Delta B}{B^2}$</td><td>$\frac{\Delta A}{A} + \frac{\Delta B}{B}$</td><td>$\left(\frac{\Delta A}{A} + \frac{\Delta B}{B}\right) \times 100$</td></tr>
<tr><td>Power</td><td>A^n</td><td>$nA^{n-1}\Delta A$</td><td>$n\frac{\Delta A}{A}$</td><td>$n\frac{\Delta A}{A} \times 100$</td></tr>
</table>
</td><td>A, B = the values for different physical quantities.</td></tr>
<tr><td>21.</td><td>Combination of errors (Percentage error)</td><td>General rule: If $Z = \frac{A^p B^q}{C^r}$, then the maximum fractional or relative error in Z will be,
$\frac{\Delta Z}{Z} = p\frac{\Delta A}{A} + q\frac{\Delta B}{B} + r\frac{\Delta C}{C}$
% Error in $Z = \frac{\Delta Z}{Z} \times 100$
$= p\frac{\Delta A}{A} \times 100 + q\frac{\Delta B}{B} \times 100 + r\frac{\Delta C}{C} \times 100.$</td><td>$p$, q, and r = powers of those variables.</td></tr>
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